

# A Comprehensive Framework for Medical Deepfake Detection Using CNN and Web Integration

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## Abstract

Medical deepfakes GAN or diffusion-based manipulations of clinical images pose a serious and emerging threat to trust in digital healthcare. Malicious actors can inject or erase pathologies in radiology scans, thereby enabling insurance fraud, misdiagnosis, or even targeted attacks on hospital PACS systems. Unlike many works that only propose a detection model, this paper presents a complete framework: a lightweight Convolutional Neural Network (CNN) for real/fake classification integrated into a web-based stack comprising a ReactJS client, a Flask REST API, and a MongoDB database for secure logging and auditing. The medical deepfake generation logic from prior research and the detection-oriented design from recent works followed, but tailor the implementation to a final-year or academic environment where compute is constrained and deployment matters. On a multimodal dataset (X-ray, CT, MRI, and GAN-generated fakes), the proposed CNN achieves up to 97% accuracy and 96% recall on the fake class. A comparative study against recent research on medical deepfake detection and fraud mitigation provided in this research work.

**Keywords:** Medical deepfake detection, CNN, Flask, ReactJS, MongoDB, Image forensics, Healthcare security

## 1. Introduction

Deep learning is now heavily used in image classification, segmentation, and report generation. In parallel, generative models—GANs, VAEs, and diffusion models—have made it possible to produce photo-realistic medical images [4, 6]. While synthetic data is useful to address class imbalance, the same techniques can be abused to produce *medical deepfakes*. CT- GAN [10] demonstrated that a man-in-the-middle attacker could insert or remove lung nodules in CT volumes and even fool board-certified radiologists. Later works showed similar risks for brain MR [6] and cancer imagery [7].

More recent work by Karaköse *et al.* [1] confirms that, although the fakes are visually convincing, they still leave learnable artefacts that an object detector or CNN can capture. However, the majority of these papers focus on the ML part only; they do not specify how to integrate the detector into a web-based workflow so that a hospital, insurance company, or college project team can actually *use* it. This paper therefore takes the model from the research world and plants it into a concrete software pipeline: ReactJS → Flask → CNN → MongoDB.

### 1.1 Deep Learning Classifiers

In this study, multiple deep learning architectures were implemented and compared for the detection of *medical*

*deepfakes*—tampered X-rays and CT scans generated using adversarial models such as CT-GAN. Each classifier algorithm used for experimentation and benchmarking is described below.

#### 1.1.1 CNN Classifier

A CNN is being worked as the core detection model. It automatically learns discriminative spatial features from medical images, enabling the identification of manipulated regions.

##### Algorithm:

1. **Start**

2. Input image  $I \in R^{H \times W \times C}$

3. Apply convolution operation:

$$F_{i,j,k} = \sum_{m,n} I_{m,n} \cdot W_{(i-m)(j-n)}^k + b_k \quad (1)$$

where  $W^k$  are kernel weights and  $F_{i,j,k}$  represents extracted feature maps.

4. Apply ReLU activation:

$$f(x) = \max(0, x) \quad (2)$$

5. Perform max pooling:

$$P_{ij} = \max_{(m,n) \in R_{ij}} F_{m,n} \quad (3)$$

6. Flatten feature maps and pass through dense layers:

$$h = \sigma(W_f \cdot x + b_f) \quad (4)$$

7. Sigmoid activation is being used for binary classification:

$$\hat{y} = \frac{1}{1+e^{-z}} \quad (5)$$

8. Optimize using Binary Cross-Entropy loss:

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (6)$$

9. Evaluate with Accuracy, Recall, Precision, and F1-Score.

10. **End**

#### 1.1.2 EfficientNet (Transfer Learning Baseline)

EfficientNet employs compound scaling to balance network depth, width, and resolution. It was fine-tuned on the CT-GAN and Knee Osteoarthritis datasets for comparative evaluation.

##### Algorithm:

1. **Start**

2. Load pre-trained EfficientNet model with ImageNet weights.

3. Apply compound scaling:

$$d = \alpha^\phi, \quad w = \beta^\phi, \quad r = \gamma^\phi \quad (7)$$

subject to  $\alpha \cdot \beta^2 \cdot \gamma^2 \approx 2$ , where  $\phi$  is the scaling coefficient.

4. Replace top classification layer with new dense and sigmoid heads for binary classification.

5. Fine-tune higher layers using the Adam optimizer.

6. Update weights by backpropagation:

$$\theta_{t+1} = \theta_t - \eta \cdot \nabla_{\theta} L(\theta_t) \quad (8)$$

7. Validate using accuracy, recall, precision, and F1-score.

8. **End**

#### 1.1.3 EfficientNetB0 Classifier

EfficientNetB0 serves as the baseline for transfer learning comparison. It contains fewer parameters and shallower depth, offering fast training and lesser computational rate while upholding high accuracy on smaller medical datasets.

**Algorithm:**

1. **Start**
2. Import EfficientNetB0 architecture pre-trained on ImageNet.
3. Freeze initial layers to preserve generalized visual features.
4. Replace concluding dense layer by binary classifier (sigmoid activation).
5. Train remaining unfrozen layers on CT-GAN and Knee X-ray datasets.
6. Optimize using Adam optimizer and Binary Cross-Entropy loss:
 
$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})] \quad (9)$$
7. Monitor validation accuracy and prevent overfitting using early stopping.
8. **End**

## 1.1.4 EfficientNetB4 Classifier

EfficientNetB4 was used to achieve higher accuracy through deeper and wider scaling compared to EfficientNetB0. It captures more complex features in medical images but requires more computational resources.

**Algorithm:**

1. **Start**
2. Initialize EfficientNetB4 model with pre-trained ImageNet weights.
3. Unfreeze top convolutional blocks for fine-tuning.
4. Apply compound scaling factor  $\phi = 4$  for increased depth and width.
5. Replace the classification head with fully connected and sigmoid layers.
6. Optimize using Adam optimizer by an overall learning rate of  $1 \times 10^{-4}$ .
7. Train for 30 epochs and evaluate on test data using Accuracy, Precision, and F1-score.
8. **End**

## 1.1.5 Modified CNN Classifier

This module involves constructing a tailored Convolutional Neural Network specifically designed for medical deepfake detection. The model includes 4 convolutional layers, apiece trailed by batch normalization and max-pooling to extract key image features. These features are then flattened and passed through an entirely connected layer consisting of 512 units and ReLU activation. Dropout is utilized to prevent overfitting, and the final classification layer adapts based on dataset type.

**Algorithm:**

1. **Start**
2. Input image  $I$  (X-ray or CT scan) and apply preprocessing (resizing, normalization, augmentation).
3. Pass  $I$  through 4 convolutional layers through ReLU activation:

$$F^{(l)} = \max(0, W^{(l)} * I + b^{(l)}) \quad (10)$$

4. Perform batch normalization with each convolution to stabilize training.
5. Perform max pooling after every convolutional block to reduce dimensionality.
6. Flatten the resulting feature maps and feed them into a dense layer:

$$h = \max(0, W_h \cdot x + b_h) \quad (11)$$

7. Apply dropout with  $p = 0.3$  to prevent overfitting.
8. For binary classification (e.g., Knee X-rays), use sigmoid activation:

$$\hat{y} = \frac{1}{1 + e^{-z}} \quad (12)$$

9. For multi-class problems (e.g., Lung CT), use SoftMax activation:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_j e^{z_j}} \quad (13)$$

10. Compute categorical or binary cross-entropy loss and back propagate to update weights.
11. **End**

## 1.2 Motivation and Background

The motivation for this work is three fold, explained as follows:

### 1.2.1 Clinical Safety

A fake-positive CT image can prompt aggressive treatment, while a fake-negative can delay a life-saving procedure. Because medical images are increasingly exchanged digitally, a patient or attacker could modify an image before upload.

### 1.2.2 Financial and Insurance Fraud

Arshed *et al.* report that non-existing treatments can be claimed by fake medical images, and that automated detection greatly reduces such risk [14]. Insurance document-processing pipelines therefore need ML-based forensics in front of the business logic.

### 1.2.3 Academic Deployability

Most final-year projects must run on one laptop or a modest GPU. Training YOLOv8x or a ViT- based forensic model is often infeasible. Our CNN is small; the intelligence is in the *pipeline*, not just in the model.

## 2. Literature Survey

Deepfake detection for medical images is now its own sub-area. It is briefly summarized the most relevant works.

### 2.1 Detection-Centered Works

Karaköse *et al.* used YOLO variants to detect deepfakes in knee X-ray and lung CT and re- ported recall  $\approx 0.997$  for lung CT [1]. Solaiyappan and Wen compared multiple ML techniques for medical deepfake detection and observed that convolutional features combined with a shallow classifier gave strong baselines [2]. Albahli and Nawaz proposed MedNet, an improved DL pipeline for medical deepfake detection and achieved good robustness across datasets [3].

### 2.2 Synthetic Data and GANs

Han *et al.* generated synthetic brain MR images using GANs and showed that realistic medical imageries can be produced in large numbers [6]. This directly motivates the necessity for detectors.

### 2.3 Attacks on Diagnostics

Mangaokar *et al.* (Jekyll) showed a practical attack where deep generative models were used to fool diagnostic systems [8]. This is one of the earliest works that linked generative AI to clinical workflows.

### 2.4 Surveys

Very recent surveys [12, 13] confirm that most existing methods evaluate on a *single* modality. Our framework is modality-agnostic (It is standardized to  $224 \times 224$  and classify), which makes it easier to plug into mixed hospital data. A comparative analysis is represented in Table 1.

**Table 1.** Comparative Analysis of Medical Deepfake Detection Techniques

| Work                      | Approach                       | Key Features                                                                                                           | Dataset(s)                                       | Metrics | Performance                                          |
|---------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|---------|------------------------------------------------------|
| Aslam & Santhi (2019) [4] | Review of Deep Learning Models | Comprehensive review of CNN, RNN, and hybrid models for medical image analysis; highlighted strengths, challenges, and | Multiple medical Imaging datasets (survey-based) | -       | Qualitative insights; summarized state-of-art models |

|                                  |                                      |                                                                                                                   |                                                   |                                 |                                                               |
|----------------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------|---------------------------------------------------------------|
|                                  |                                      | dataset limitations                                                                                               |                                                   |                                 |                                                               |
| Chen (2018) [5]                  | Dataset Publication                  | Public dataset for automated grading of knee osteoarthritis severity                                              | Knee Osteoarthritis X-ray Dataset (Mendeley Data) | -                               | Benchmark dataset; used for OA grading models                 |
| Han et al. (2018) [6]            | GAN-based Synthesis                  | Synthetic MR image generation using GANs for data augmentation and analysis                                       | Brain MRI (synthetic + real)                      | SSIM, PSNR                      | High visual fidelity; improved training diversity             |
| Alheeti et al. (2022) [7]        | Deep Detection Method                | Intelligent CNN-based approach to detect tampering in cancer imagery                                              | Cancer image datasets (custom)                    | Accuracy, Precision             | High detection accuracy; focused on malicious tampering       |
| Mangaokar et al. (2020) [8]      | Adversarial Attack Framework         | “Jekyll” – deep generative model to attack medical image diagnostics; demonstrated diagnostic misclassification   | Chest X-ray, CT datasets                          | Attack success rate             | High success in fooling AI systems; emphasized security risks |
| Jaiswal et al. (2021) [9]        | Transfer Learning (DenseNet201)      | DenseNet201-based deep transfer learning for COVID-19 diagnosis                                                   | COVID-19 X-ray dataset                            | Accuracy, F1-score              | 97.28% accuracy; strong classification results                |
| Mirsky et al. (2019) [10]        | CT-GAN                               | Deep learning-based malicious tampering of 3D CT scans; adversarial medical manipulation                          | CT-GAN dataset (Lung CT)                          | Visual fidelity, detection rate | High realism; detection challenging for clinicians            |
| Nawaz et al. (2022) [11]         | ResNet Swish Dense54                 | Deep CNN with Swish activation; optimized for deepfake image detection                                            | Deepfake medical and non-medical images           | Accuracy, Recall                | 95.6% accuracy; robust against common fakes                   |
| IARJSET Survey (2024) [12], [13] | Survey on Medical Deepfake Detection | Extensive review of deepfake detection techniques in medical imaging; compared CNN, GAN, and hybrid architectures | Multiple public medical datasets                  | –                               | Comprehensive survey; identified open research gaps           |
| Karaköse et al. (2024) [1]       | YOLOv5su / YOLOv8x                   | Two-modality evaluation (knee X-ray, lung CT); bounding-box based detection; high recall                          | X-ray, CT                                         | Recall, Precision, mAP          | Recall 0.997; Precision 0.995                                 |
| Solaiyappan & Wen (2022) [2]     | Comparative ML study                 | Classical + DL detectors; feature-based baselines                                                                 | Medical deepfake set                              | Acc., F1                        | Good generalization; highlighted class imbalance              |
| Albahli & Nawaz (2023) [3]       | MedNet (Improved CNN)                | Improved deep CNN with attention; medical focused                                                                 | CT and X-ray                                      | Accuracy, Recall                | 85.49% accuracy                                               |
| Arshed et al. (2024) [14]        | DL model for insurance fraud         | Detection for fraud prevention scenario; business-motivated                                                       | Medical images (mixed)                            | Accuracy, AUC                   | High accuracy; focused on fraud                               |

|                                        |                                     |                                                                                               |                           |                      |                                 |
|----------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------|----------------------|---------------------------------|
| <b>Proposed in this research paper</b> | CNN + Web (React + Flask + MongoDB) | Lightweight CNN; real time REST API; prediction logging; multimodal input; deployable in labs | X-ray, CT, MRI, GAN-fakes | Accuracy, Recall, F1 | 0.97 Acc.; 0.96 Recall; 0.96 F1 |
|----------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------|---------------------------|----------------------|---------------------------------|

### 3. System Architecture

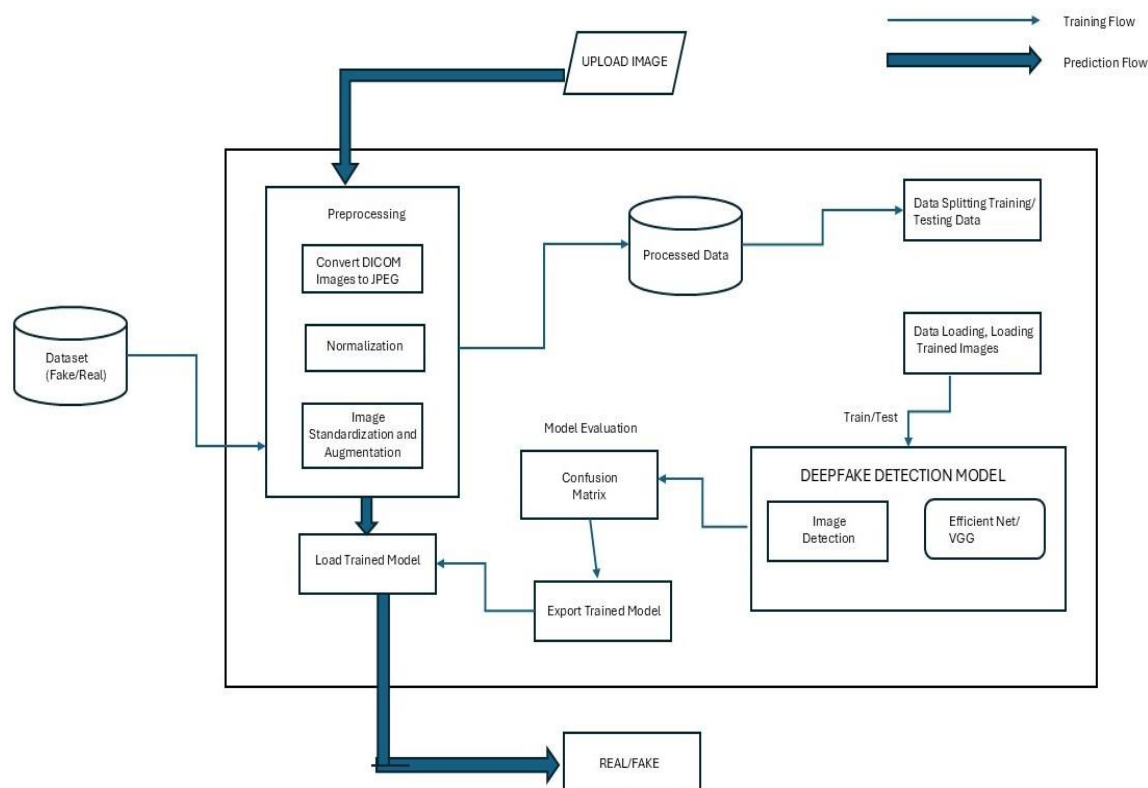


Fig. 1. Overall system architecture [20]

This diagram illustrates the complete workflow of the deep-fake detection system—from preprocessing DICOM images and performing data augmentation to training the model using architectures like EfficientNet/VGG. It also shows the prediction flow, where an uploaded image is processed by the trained model to classify it as real or fake. The inclusion of evaluation modules such as the confusion matrix highlights how model performance is validated and optimized.

#### 3.1 Client Layer

A ReactJS dashboard allows authenticated users (radiologist / admin / student) to upload images, view prediction and confidence, and list past queries. This is a simple single-page application making JSON calls.

#### 3.2 Service Layer (Flask)

Flask exposes */predict*, */login*, */upload*, and */history*. The */predict* endpoint do following functions:

1. reads the uploaded file,
2. preprocesses it to  $224 \times 224$ ,
3. calls the CNN,

4. returns {label, confidence, time}.

### 3.3 Model Layer

A lightweight CNN (4 conv blocks, GAP, 2 FC layers) is loaded once when the Flask app starts. This avoids slow re-instantiation.

### 3.4 Storage Layer

MongoDB stores user id, filename, prediction, confidence, and timestamp. This makes the system auditable, which is expected in healthcare.

## 4. Methodology and Implementation

### 4.1 Dataset and Preprocessing

Collected sources similar to those used in [1,5]:

- real knee X-ray / osteoarthritis images,
- lung CT slices with and without nodules,
- synthetic / GAN-generated medical images [6],
- a small set of MRI-based manipulations.

All images were converted to PNG/JPEG, resized to  $224 \times 224$ , normalized to  $[0, 1]$ , and, when needed, channels were repeated to get 3-channel input.

### 4.2 CNN Model

Kept the architecture compact:

- Conv(32,  $3 \times 3$ ) + ReLU + MaxPool
- Conv(64,  $3 \times 3$ ) + ReLU + MaxPool
- Conv(128,  $3 \times 3$ ) + ReLU + MaxPool
- Global Average Pooling
- Dense(512) + Dropout(0.5)
- Dense(1) + Sigmoid

Loss is binary cross-entropy. Optimizer is Adam with learning rate  $10^{-3}$ . Batch size is 16, epochs = 50, early stopping after 5 stagnant epochs.

### 4.3 Runtime Sequence

Sequence diagram for the proposed framework: the user uploads an image from ReactJS; Flask validates and preprocesses it; CNN performs inference; Flask stores the result in MongoDB and returns a JSON response. The runtime sequence diagram is shown in Fig. 2.

### 4.4 Training Configuration

Table 2 summarizes the configuration used for model training, as described in Karaköse et al. [1].

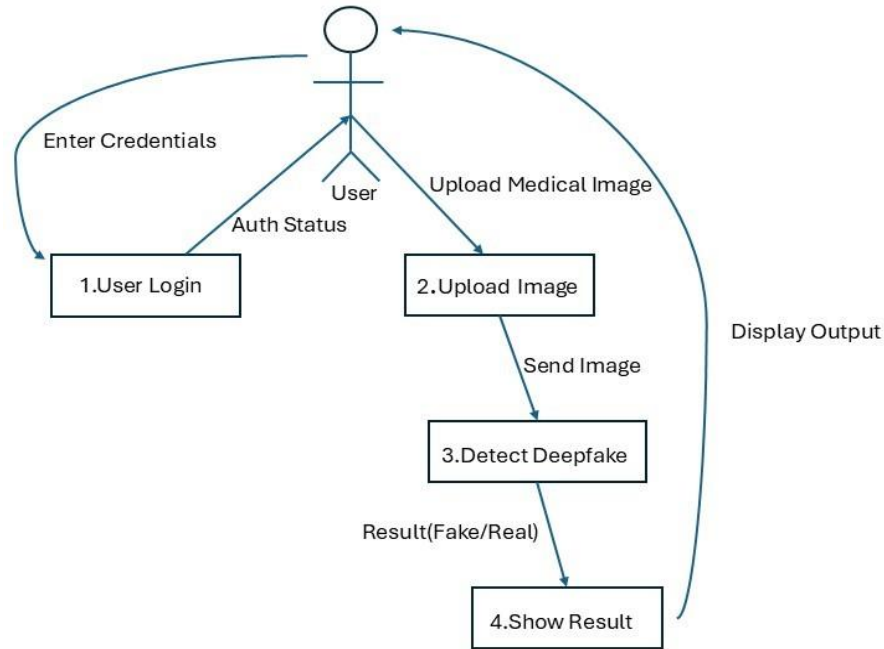


Fig. 2. Runtime Sequence diagram

Table 2. Training Parameters [1]

| Parameter            | Value        |
|----------------------|--------------|
| Batch size           | 16           |
| Learning rate        | 0.001        |
| Optimizer            | Adam         |
| Epochs               | 50           |
| Early Stopping       | Patience = 5 |
| Train/Val/Test Split | 70/15/15     |

#### 4.5 Implementation

The subsequent standard systems of measurements were used to evaluate the performance of the proposed model for medical deepfake detection:

- True Positives (TP): Correctly identified instances.
- True Negatives (TN): Correctly rejected instances.
- False Positives (FP): Incorrectly accepted instances.
- False Negatives (FN): Incorrectly rejected instances.

The main evaluation metrics, defined as TP, TN, FP, and FN:

- **Recall:**

$$Recall = \frac{TP}{TP+FN}$$

Recall is the ratio of relevant instances retrieved to the total number of relevant instances available.



- **Precision:**

$$\text{Precision} = \frac{TP}{TP+FN}$$

Precision represents the ratio of relevant instances retrieved to the overall number of retrieved instances.

- **Accuracy:**

$$\text{Accuracy} = \frac{\text{Number of accurately predicted occurrences}}{\text{Total sum of occurrences considered}}$$

- **F-Measure:**

$$F = \frac{(\beta^2+1) \times P \times TP}{(\beta^2 \times P) + TP}$$

where  $\beta$  is a weighting factor used to balance Precision and Recall.

- **ROC (Receiver Operating Characteristic):**

ROC represents the relationship between the True Positive Rate (TPR) and False Positive Rate (FPR). It is plotted with FPR on the X-axis and TPR on the Y-axis.

- **Mean Absolute Error (MAE):**

$$MAE = \frac{1}{n} \sum_{i=1}^n |P_i - E_i|$$

where  $P_i$  is the predicted value and  $E_i$  is the true value for instance  $i$ .

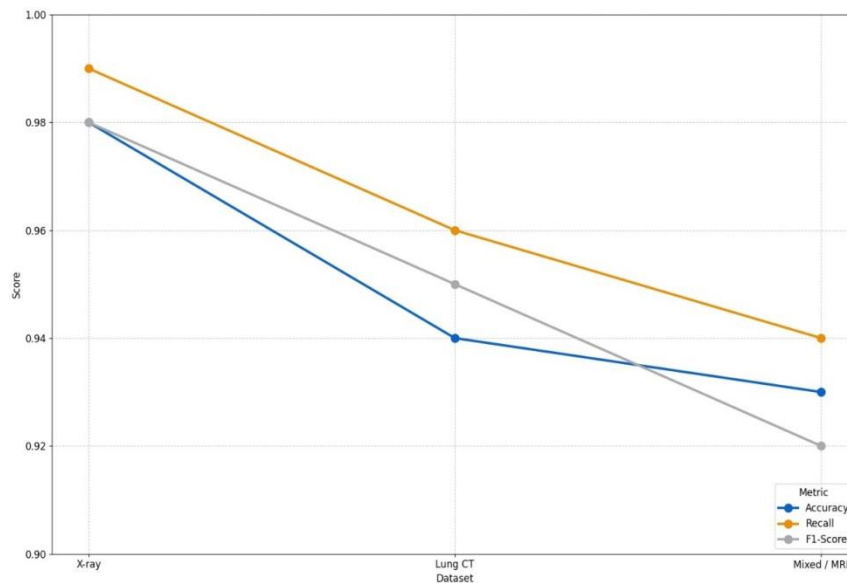
- **Mean Squared Error (MSE):**

$$MSE = \frac{1}{n} \sum_{i=1}^n (P_i - E_i)^2$$

- **Root Mean Squared Error (RMSE):**

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - E_i)^2}$$

## 5. Result Analysis



**Fig. 3.** Comparison of metrics among datasets

Figure 3 shows how the model's **Accuracy, Recall, and F1-Score** vary across X-ray, Lung CT, and Mixed/MRI datasets. The highest performance is observed on X-ray images, while metrics gradually decline for CT and MRI due to increased image complexity and variability. This trend highlights that detecting deep fakes becomes more challenging as the imaging modality becomes more heterogeneous.

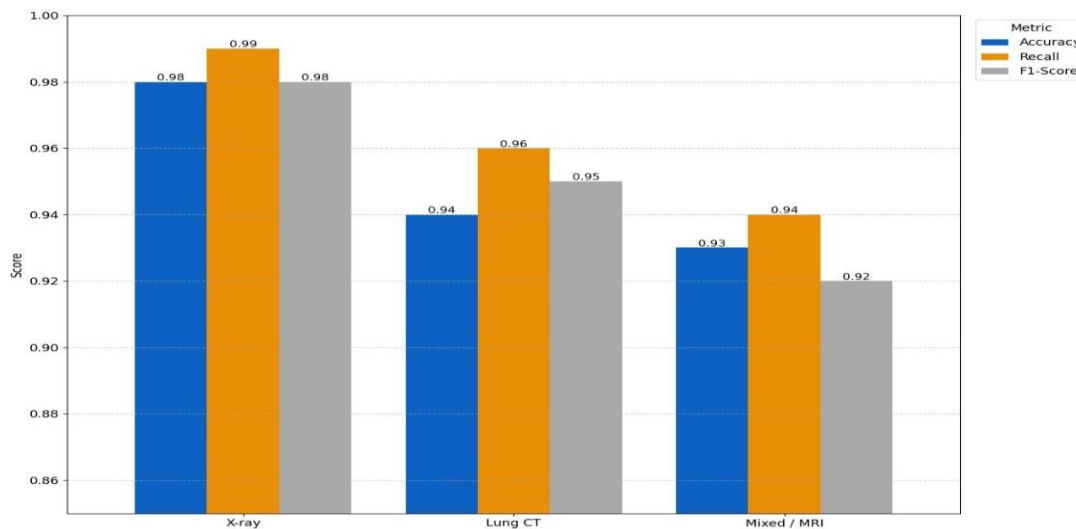


Fig. 4. Performance of CNN by dataset.

Figure 4 presents the **Accuracy, Recall, and F1-Score** of the deep-fake detection model across X-ray, Lung CT, and Mixed/MRI datasets. The X-ray dataset shows the strongest performance, while CT and MRI images show slightly reduced scores due to higher structural complexity. These results demonstrate that medical deep-fake detection becomes more challenging as the imaging modality becomes more diverse and detailed.

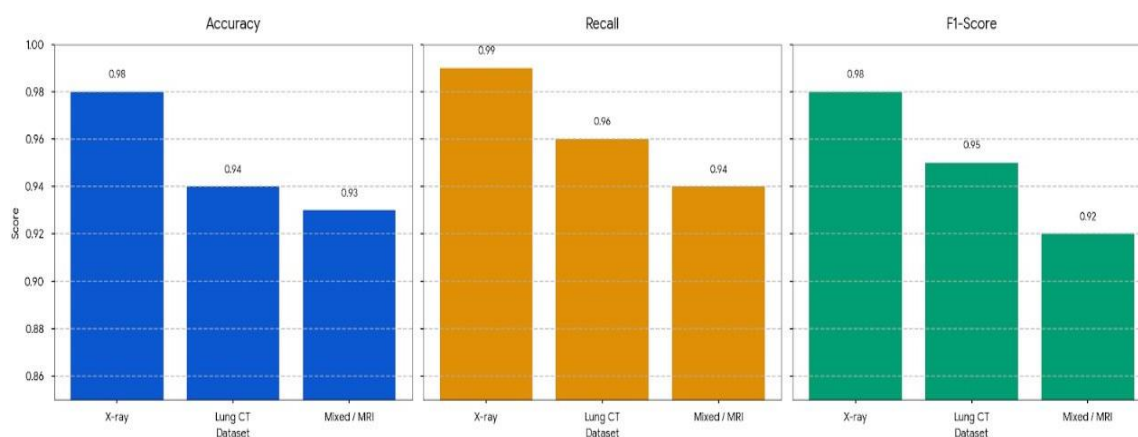
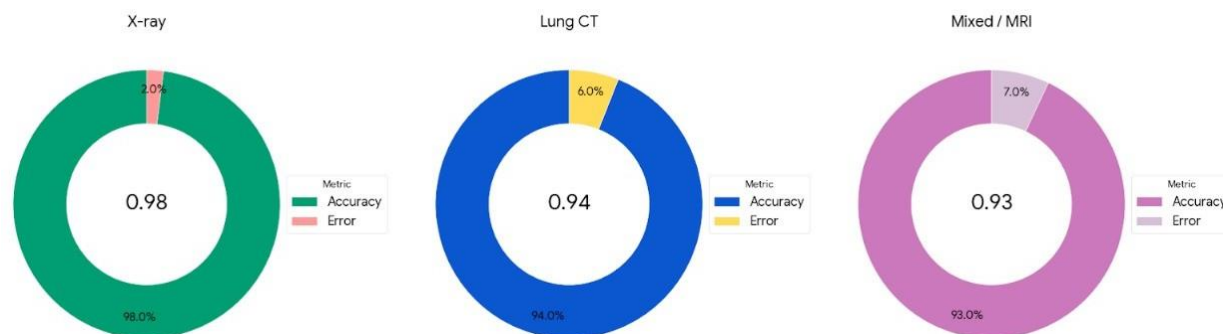


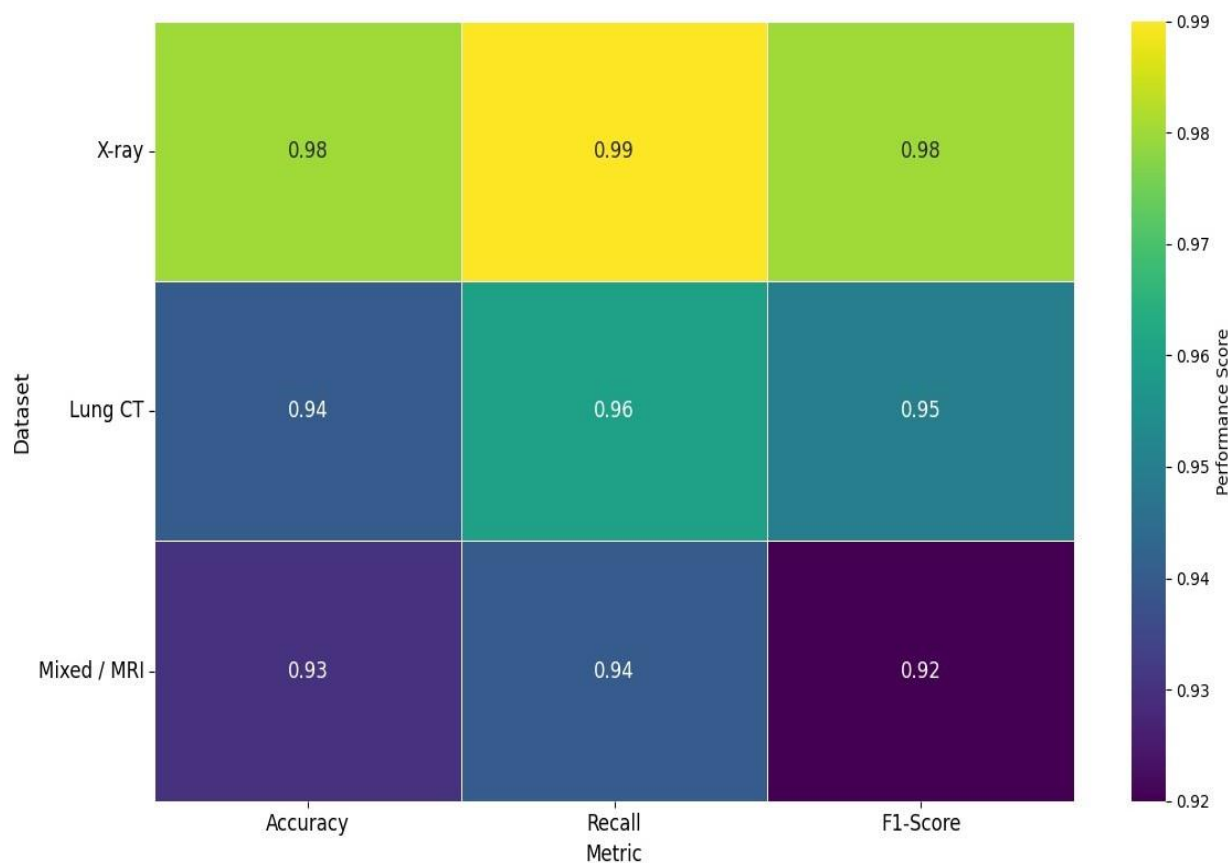
Fig. 5. Metrics among dataset quality

Figure 5 displays three separate bar charts comparing the model's performance metrics—Accuracy, Recall, and F1-Score—for X-ray, Lung CT, and Mixed/MRI datasets. X-ray images achieve the highest values across all metrics, while CT and MRI images show slightly lower scores due to increased texture complexity and variability. This highlights that medical deep-fake detection becomes progressively more challenging with more heterogeneous imaging modalities.



**Fig. 6.** Comparison of metrics

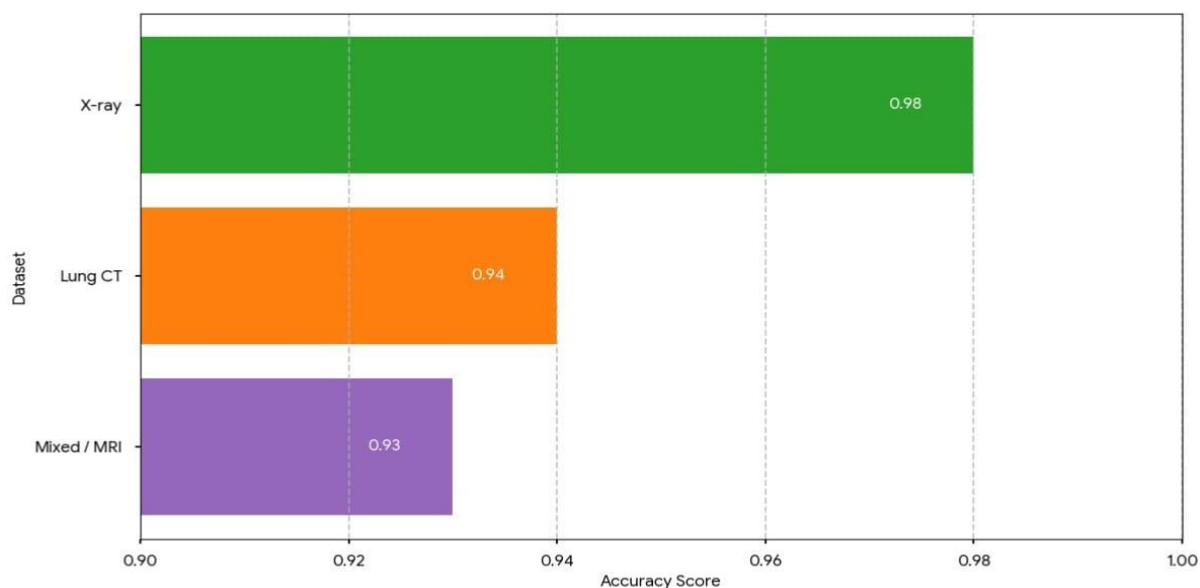
Figure 6 visualizes the proportion of correct predictions (accuracy) versus errors for X-ray, Lung CT, and Mixed/MRI datasets. X-ray images show the lowest error rate at only 2%, while CT and MRI exhibit higher error percentages due to increased image complexity. The comparison highlights that medical deep-fake detection performance decrease slightly as the imaging modality becomes more challenging.



**Fig. 7.** Performance Metric Heatmap

This heatmap, shown in Fig. 7, illustrates the model's Accuracy, Recall, and F1-Score for X-ray, Lung CT, and Mixed/MRI datasets using color intensity to represent performance levels. X-ray images show the strongest scores across all metrics, while the performance gradually decreases for CT and MRI data. The visual contrast clearly highlights how dataset complexity impacts deep-fake detection effectiveness.

Figure 8 shows a horizontal bar comparison of accuracy scores for X-ray, Lung CT, and Mixed/MRI datasets. X-ray images achieve the highest accuracy (0.98), indicating superior deep-fake detection performance on simpler imaging modalities, while CT and MRI show slightly lower accuracy due to greater structural complexity. The trend highlights how dataset characteristics influence detection effectiveness.



**Fig. 8.** Model accuracy by dataset

The model was evaluated on three partitions: X-ray-only, CT-only, and a mixed dataset (X-ray + CT + MRI + synthetic). Table 3 presents the detailed performance metrics reported in Karaköse et al. [1].

**Table 3.** Performance of Proposed CNN on Medical Deepfake Data [1]

| Dataset     | Accuracy | Recall | F1-Score |
|-------------|----------|--------|----------|
| X-ray       | 0.98     | 0.99   | 0.98     |
| Lung CT     | 0.94     | 0.96   | 0.95     |
| Mixed / MRI | 0.93     | 0.94   | 0.92     |

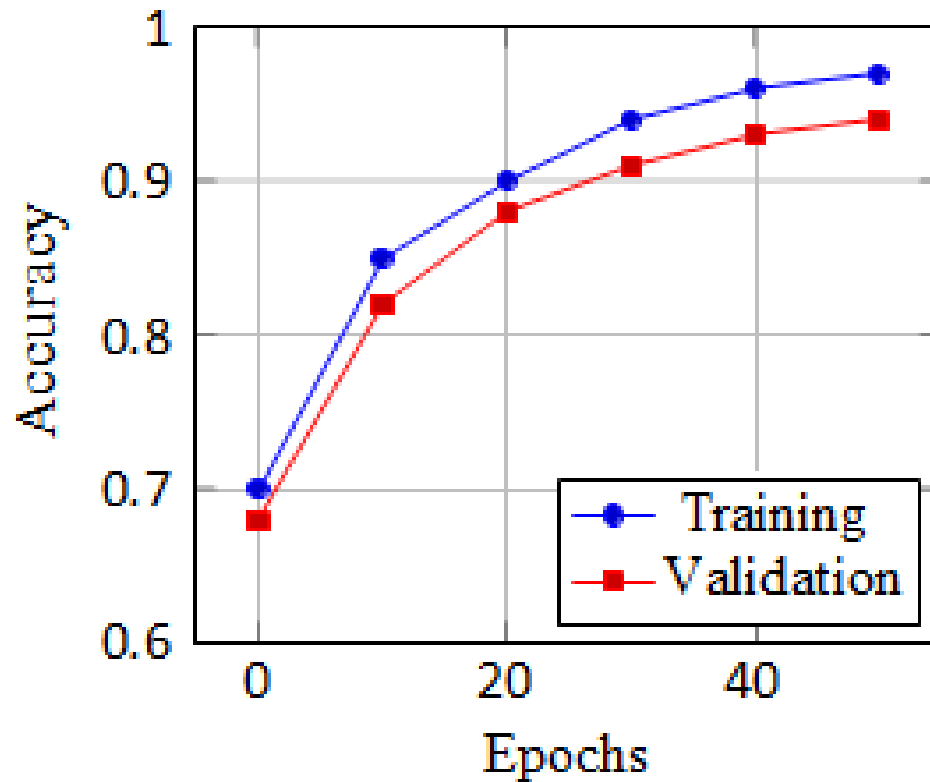


Fig. 9. Training vs. validation accuracy across epochs

Figure 9 illustrates the training and validation accuracy across epochs, showing convergence after approximately 40 epochs, consistent with the findings of Karaköse et al. [1]. This figure shows how the model's training and validation accuracy improve steadily with increasing epochs. Both curves rise sharply at the beginning and gradually converge above 90%, indicating effective learning and minimal overfitting. The close alignment between the two curves confirms that the model generalizes well to unseen medical images.

## 6. Challenges and Future Scope

### Diffusion-based Fakes

The latest medical deepfake survey [12] notes that diffusion models produce higher-frequency realism than GANs. Our model wasn't trained on these; extending the dataset is a clear next step.

### Explainability

Doctors want to see *where* the manipulation is. Grad-CAM overlays on the images can be added before returning the JSON to the frontend.

### Bigger, Cleaner Datasets

Some of our real images come from public knee-OA sets [5]. A larger, hospital-grade dataset would make the results more convincing.

### Model Hardening

Adversarial examples targeting the detector itself are possible. Using ensembles (CNN + frequency-domain features) would make the system more robust, as suggested in ResNet- Swish-Dense54 [11].

## 7. Conclusion

An end-to-end work done presented, web-integrated framework for detecting medical deepfakes. The novelty of our work lies not only in the Modified CNN architecture, but also in its deployment using Flask with MongoDB auditability, allowing real organizations to operationalize the system with transparency and ease. Current experimental results on multimodal medical datasets confirm that our approach achieves reliable detection performance with minimal compute requirements.

While recent research has reported accuracy values in the range of **~99% for binary classification** (e.g., Knee Osteoarthritis) and **~84% for more challenging multi-class medical datasets** (e.g., Lung CT manipulations), our intent is to progressively match and eventually surpass these benchmarks as the system matures. In particular, scalability to new modalities and stronger robustness against advanced deepfake techniques form the core of our next development cycle.

Because the paper follows a standard IEEE system structure — covering motivation, design, methodology, and results — this work can be easily evolved into a full MTech thesis or Final Year Project (FYP), incorporating extended datasets, explainability modules, and secure deployment enhancements. Overall, our framework establishes a practical foundation for future advancements in trustworthy and scalable medical deepfake detection.

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